

5.3.16 Temperature sensor characteristics

Table 57. Temperature sensor characteristics

Symbol	Parameter	Min	Typ	Max	Unit
$T_L^{(1)}$	V_{SENSE} linearity with temperature	-	± 1	± 5	$^{\circ}\text{C}$
Avg_Slope ⁽²⁾	Average slope from V_{SENSE} voltage	2.4	2.53	2.65	mV/ $^{\circ}\text{C}$
$V_{30}^{(3)}$	Voltage at 30 $^{\circ}\text{C}$ (± 5 $^{\circ}\text{C}$)	0.742	0.76	0.786	V
$t_{START(TS_BUF)}^{(1)}$	Sensor Buffer Start-up time in continuous mode	-	8	15	μs
$t_{START}^{(1)}$	Start-up time when entering in continuous mode	-	8	120	
$t_{S_temp}^{(1)}$	ADC sampling time when reading the temperature	5	-	-	
$i_{sens}^{(1)}$	Temperature sensor consumption from VDD, when selected by ADC	-	4.7	7.0	μA

1. Specified by design – Not tested in production.

2. Evaluated by characterization – Not tested in production.

3. Measured at $V_{DDA} = 3.0 \text{ V} \pm 10 \text{ mV}$. The V_{30} ADC conversion result is stored in the TS_CAL1 byte.

5.3.17 Timer characteristics

The parameters given in the following tables are specified by design. Refer to [Section 5.3.13: I/O port characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 58. TIMx⁽¹⁾ (2) characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
$t_{res(TIM)}$	Timer resolution time	-	1	-	$t_{TIMxCLK}$
		$f_{TIMxCLK} = 48 \text{ MHz}$	20.833	-	ns
f_{EXT}	Timer external clock frequency on CH1 to CH4	-	0	$f_{TIMxCLK}/4$	MHz
Res _{TIM}	Timer resolution	TIMx	-	16	bit
$t_{COUNTER}$	16-bit counter clock period	-	1	65536	$t_{TIMxCLK}$
t_{MAX_COUNT}	Maximum possible count with 16-bit counter	-	-	65536	$t_{TIMxCLK}$

1. TIMx, is used as a general term to refer to a timer (for example, TIM1).

2. Specified by design – Not tested in production.